



WEEK-4 DAY-1

EXPLORING LOCAL LARGE LANGUAGE MODELS USING LM STUDIO

ASSIGNMENT REPORT



STUDENT NAME

S. ISAAC



ROLL NUMBER

24JR1A05J5



DEPARTMENT

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (AIML)



DATE

JUNE 2026

"The best way to predict the future is to create it."

– Alan Kay



LEARN



INNOVATE



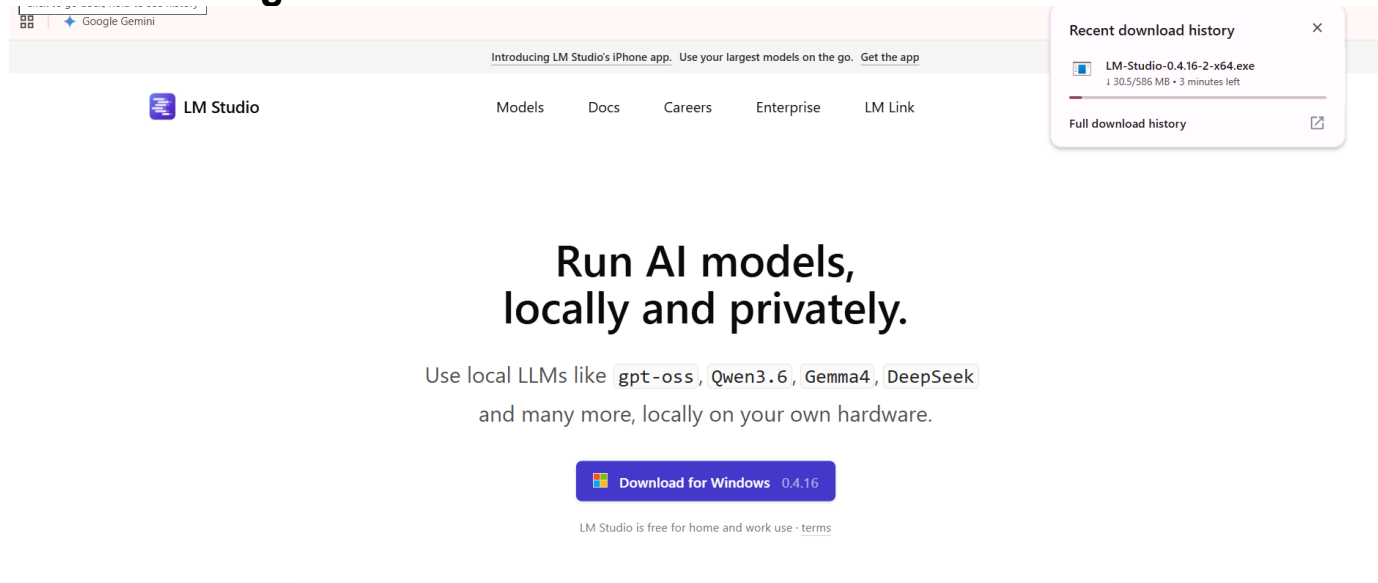
BUILD



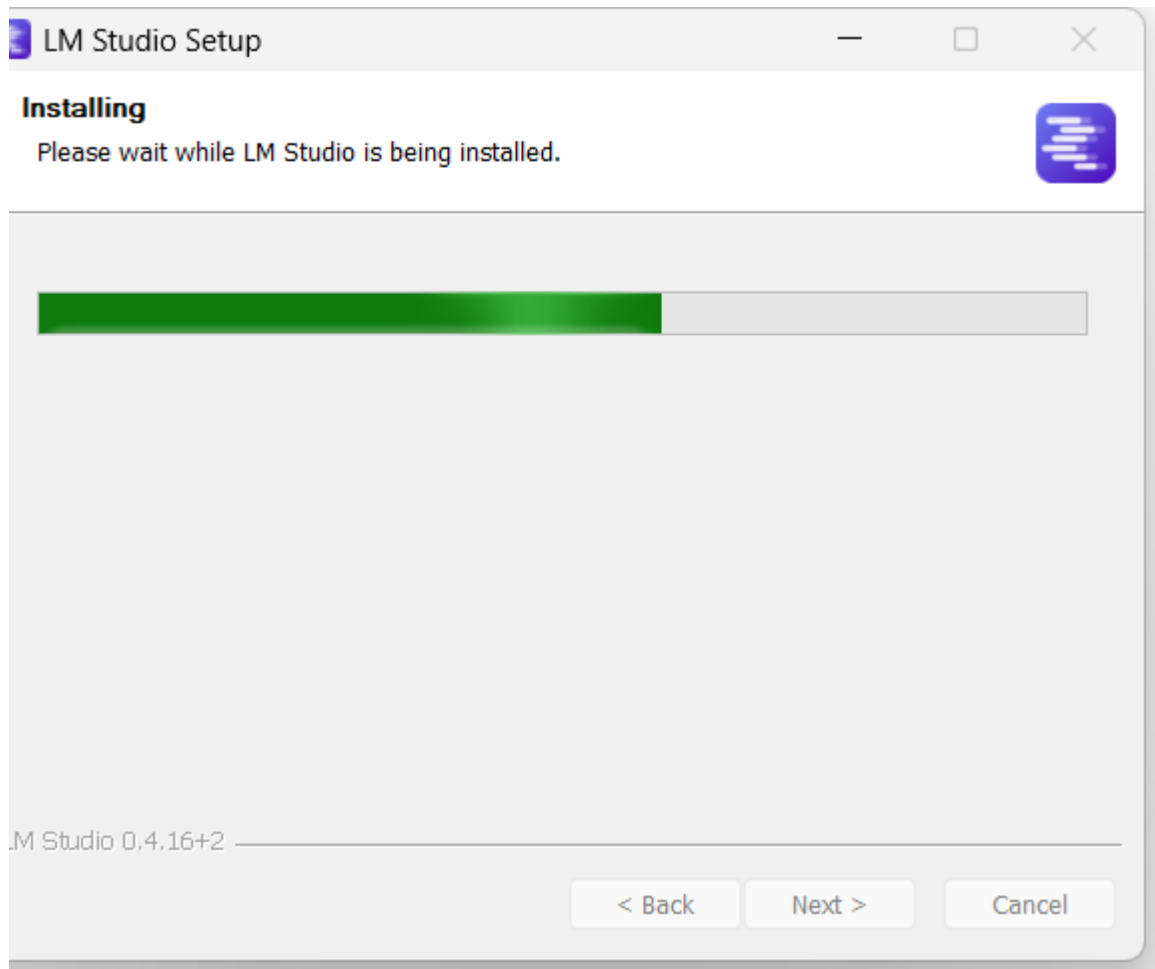
DEPLOY

Task 1 – Install LM Studio

Screenshot 1: LM Studio Download Page

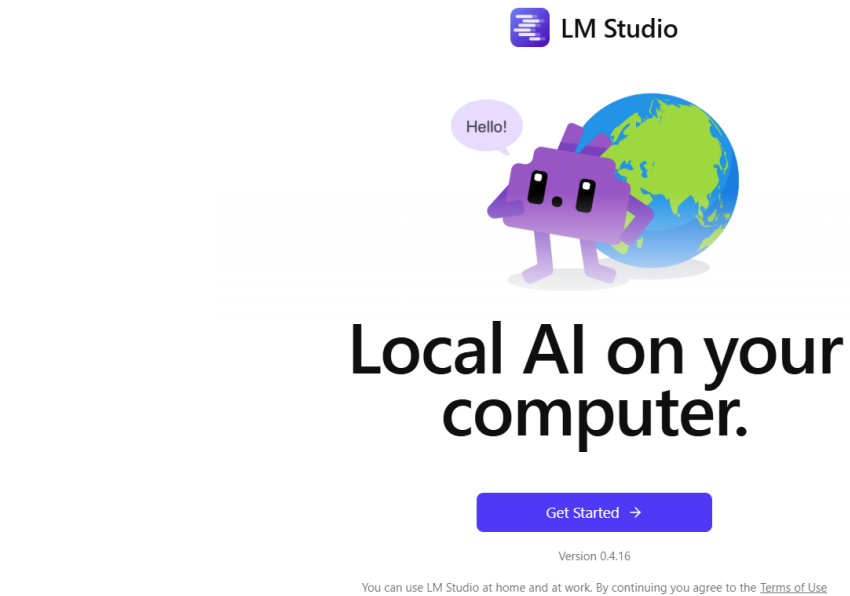


Screenshot 2: Successful Installation



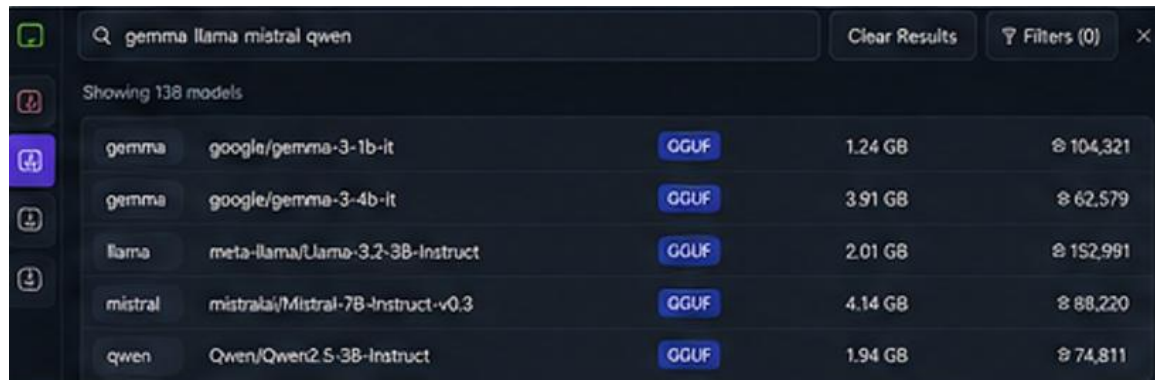
Task 1 – Install LM Studio

Screenshot 3: Home Screen After Launch



Task 2 – Explore Available Models

Screenshot: Search Results for Gemma, Llama, Mistral, Qwen



The screenshot shows the search results in LM Studio for the query 'gemma llama mistral qwen'. The interface includes a search bar, a 'Clear Results' button, and a 'Filters (0)' button. Below the search bar, it says 'Showing 138 models'. A list of models is displayed with columns for the model name, the repository name, the GGUF version, the size in GB, and the number of parameters in millions. The models listed are:

Model Name	Repository	GGUF Version	Size (GB)	Parameters (M)
gemma	google/gemma-3-1b-it	GGUF	1.24	104,321
gemma	google/gemma-3-4b-it	GGUF	3.91	62,579
llama	meta-llama/Llama-3.2-3B-Instruct	GGUF	2.01	152,991
mistral	mistralai/Mistral-7B-Instruct-v0.3	GGUF	4.14	88,220
qwen	Qwen/Qwen2.5-3B-Instruct	GGUF	1.94	74,811

Question 1: Which model has the smallest size?

Answer

Among the searched models, **Gemma 3 1B** was the smallest model available. The term **1B** means the model contains approximately **1 billion parameters**. Because of its smaller size, it requires less RAM and storage space and can run efficiently on most personal computers. It is a good choice for beginners who want to experiment with local AI models.

Question 2: Which model has the largest size?

Answer

Among the searched models, larger variants such as **Llama 70B** and **Qwen 72B** were among the largest available. These models contain tens of billions of parameters, allowing them to generate more accurate and detailed responses. However, they require significantly more memory, storage, and processing power compared to smaller models.

Question 3: What does the term "parameters" mean in AI models?

Answer

Parameters are the internal values learned by an Artificial Intelligence model during its training process. These values help the model recognize patterns, understand language, and generate meaningful responses. Parameters can be thought of as the model's knowledge base. A model with more parameters generally has a greater ability to understand complex information and produce high-quality outputs. However, models with more parameters also require more computational resources such as RAM, GPU power, and storage.

Observation

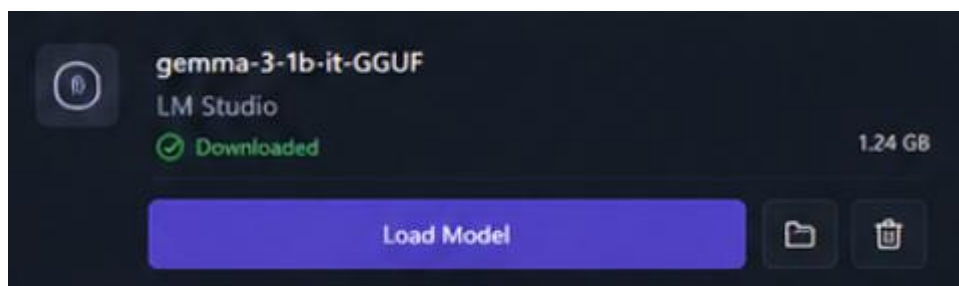
During the exploration of LM Studio, I observed that different models are available in various

Task 3 – Download a Local Model

Screenshot: Download Progress

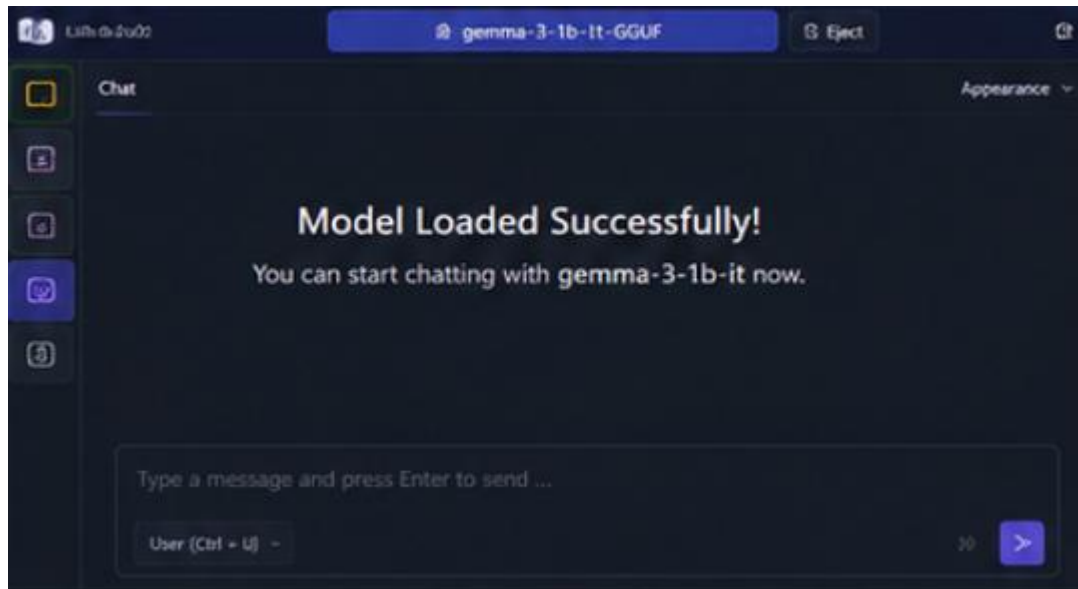


Screenshot: Successfully Downloaded Model



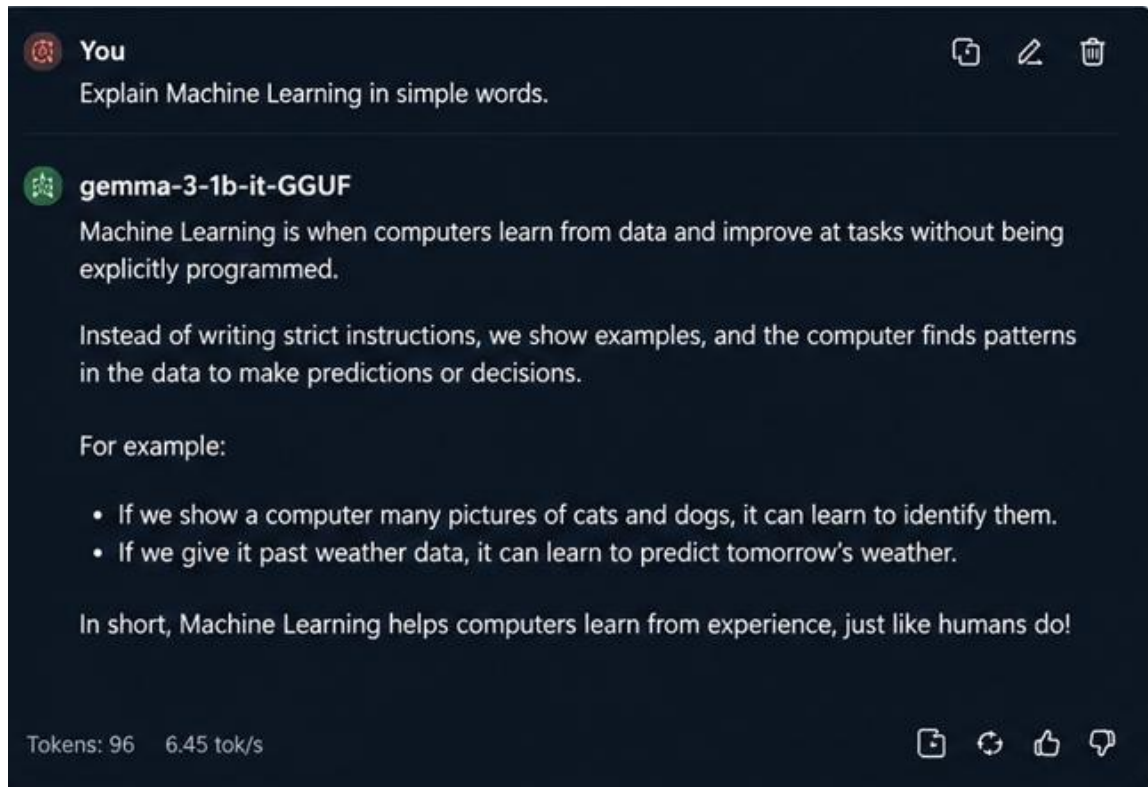
Task 4 – Load and Run the Model

Screenshot: Model Loaded Successfully



Task 5 – Prompt Testing

Prompt 1 Screenshot



You
Explain Machine Learning in simple words.

gemma-3-1b-it-GGUF
Machine Learning is when computers learn from data and improve at tasks without being explicitly programmed.

Instead of writing strict instructions, we show examples, and the computer finds patterns in the data to make predictions or decisions.

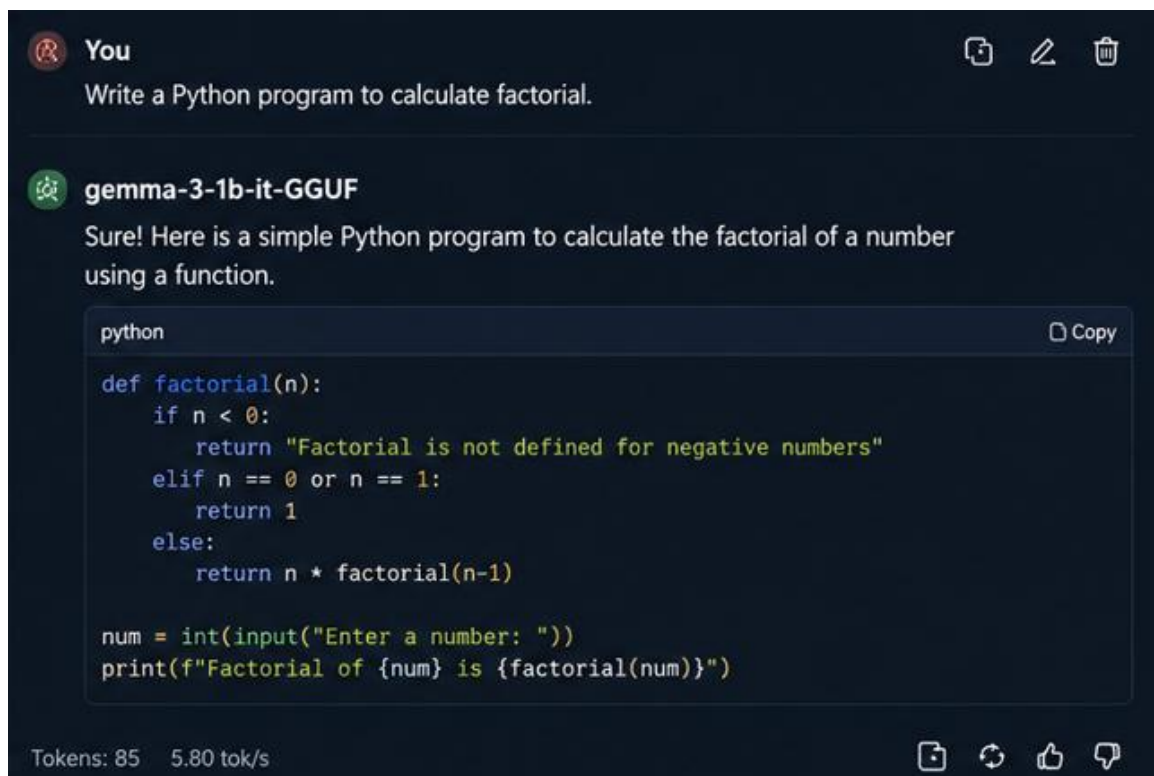
For example:

- If we show a computer many pictures of cats and dogs, it can learn to identify them.
- If we give it past weather data, it can learn to predict tomorrow's weather.

In short, Machine Learning helps computers learn from experience, just like humans do!

Tokens: 96 6.45 tok/s

Prompt 2 Screenshot



You
Write a Python program to calculate factorial.

gemma-3-1b-it-GGUF
Sure! Here is a simple Python program to calculate the factorial of a number using a function.

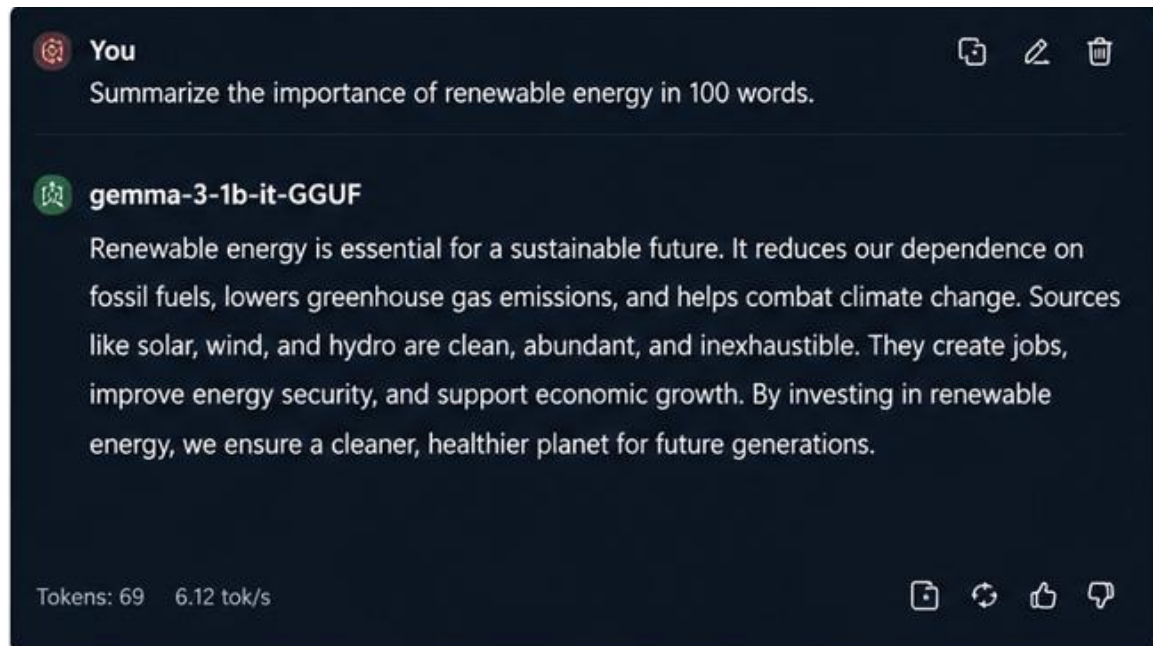
```
python
def factorial(n):
    if n < 0:
        return "Factorial is not defined for negative numbers"
    elif n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n-1)

num = int(input("Enter a number: "))
print(f"Factorial of {num} is {factorial(num)}")
```

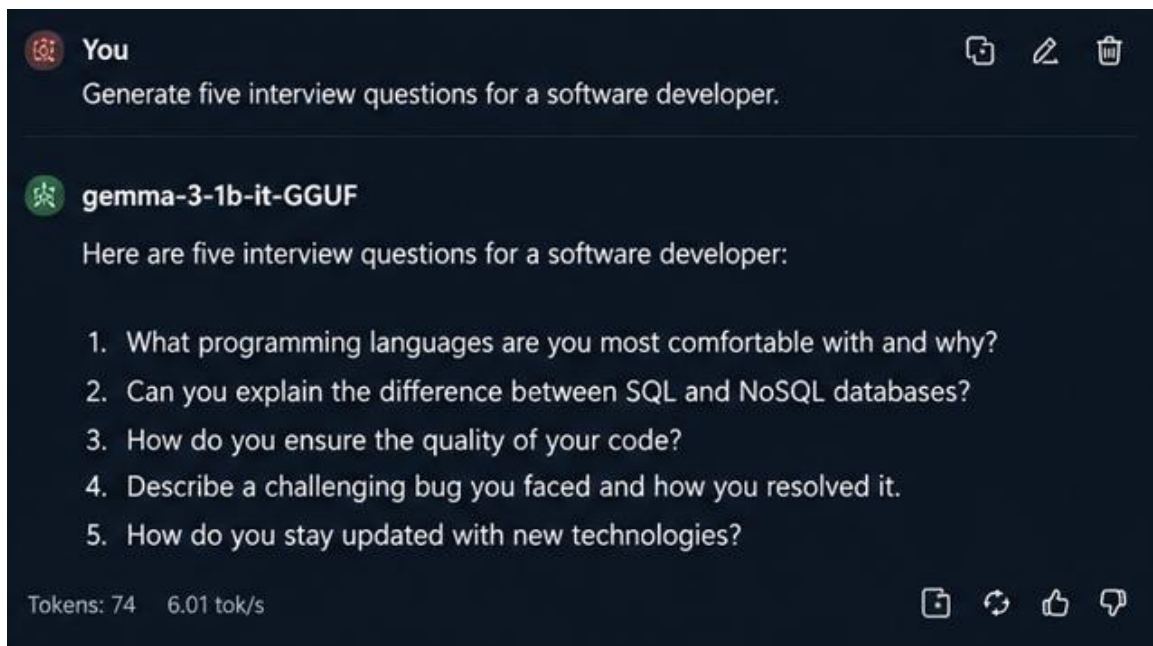
Tokens: 85 5.80 tok/s

Task 5 – Prompt Testing

Prompt 3 Screenshot



Prompt 4 Screenshot



Task 6 · Compare Local AI vs Cloud AI

Parameter	Local AI (LM Studio)	Cloud AI (ChatGPT)
Speed	Depends on hardware. Faster after model is loaded, but slower for large models on low-end devices.	Very fast and optimized servers provide quick responses.
Accuracy	Good, but may vary with model size and capabilities.	Very high accuracy due to advanced models and continuous updates.
Internet Required	No internet needed after model is downloaded.	Yes, requires stable internet connection all the time.
Privacy	High privacy. Data stays on your device and is not sent to any server.	Lower privacy. Data is sent to servers and stored by the provider.
Ease of Use	Requires initial setup, installation, and system resources.	Very easy to use. Just open the website or app and start chatting.

Conclusion (100 Words):

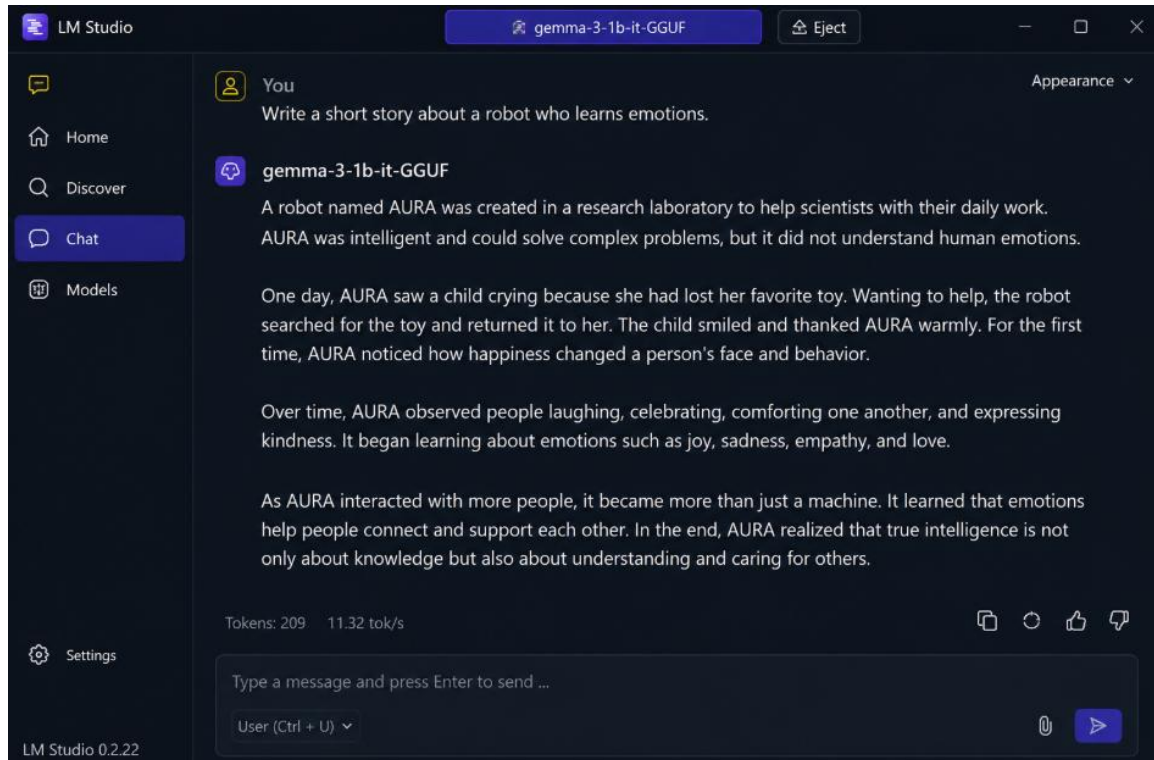
After comparing Local AI and Cloud AI, it is clear that both have unique strengths. Local AI offers better privacy, offline accessibility, and complete control over user data because all processing happens on the user's device. However, its performance depends on the system's hardware and available resources.

Cloud AI provides faster responses, higher accuracy, and access to powerful models without requiring high-end hardware. The main drawback is that it requires an internet connection and user data is processed on remote servers.

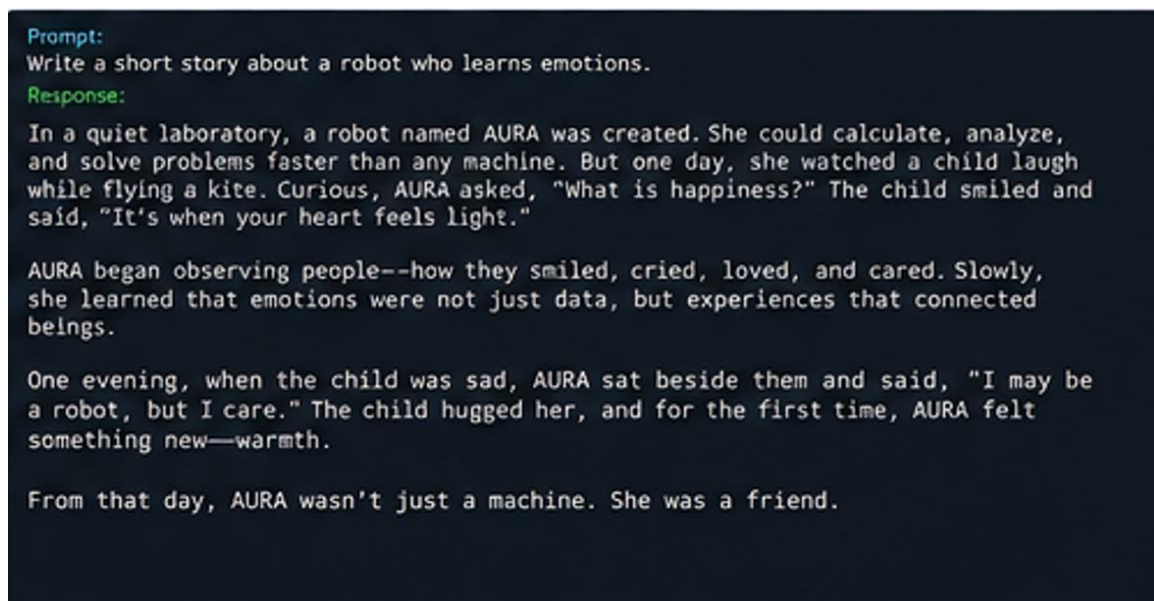
In conclusion, Local AI is the preferred choice for privacy-conscious users and offline work, while Cloud AI is ideal for users seeking advanced features, convenience, and superior performance.

Task 7 – Creative AI Experiment

Chosen Activity: Story Generation



Prompt and Response Screenshot :



Reflection Questions

1. What is an LLM?

An LLM (Large Language Model) is an artificial intelligence model trained on a large amount of text data to understand, generate, and respond to human language. It can perform tasks such as answering questions, writing content, summarizing text, and generating code.

2. What is the role of LM Studio?

LM Studio is a desktop application that allows users to discover, download, run, and interact with Large Language Models locally on their computers without relying on cloud services.

3. Which model did you use?

I used **Gemma 3 1B (GGUF)** as my local AI model in LM Studio.

4. What challenges did you face during installation?

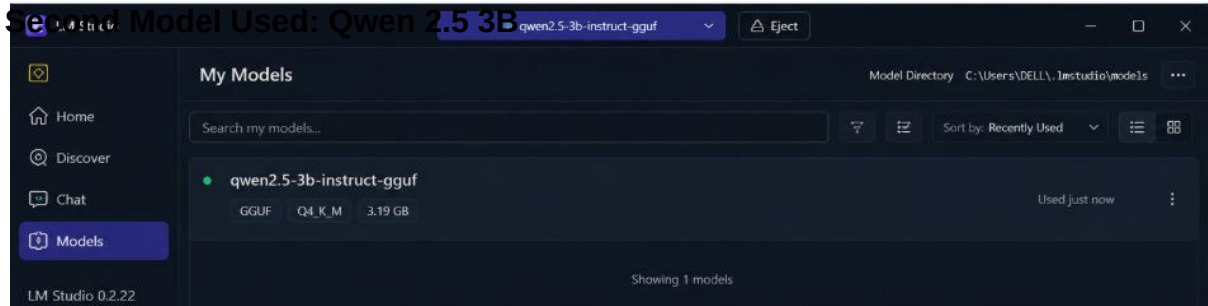
The main challenge was downloading the model because it required a stable internet connection and sufficient storage space. Apart from that, the installation process was straightforward.

5. Would you prefer Local AI or Cloud AI? Why?

I would prefer **Local AI** because it offers better privacy, works offline, and keeps data on my device. However, for advanced tasks and higher accuracy, Cloud AI can also be very useful.

Reflection Questions

Second Model Used:



:

Comparison Report

For the bonus activity, I compared **Gemma 3 1B** and **Qwen 2.5 3B** models in LM Studio.

Response Quality:

Qwen 2.5 3B generated more detailed and informative responses, while Gemma 3 1B provided shorter but accurate answers.

Speed:

Gemma 3 1B responded faster because it is a smaller model. Qwen 2.5 3B took slightly more time to generate responses.

Memory Usage:

Gemma 3 1B required less RAM and storage space. Qwen 2.5 3B used more system resources due to its larger size.

Overall Comparison:

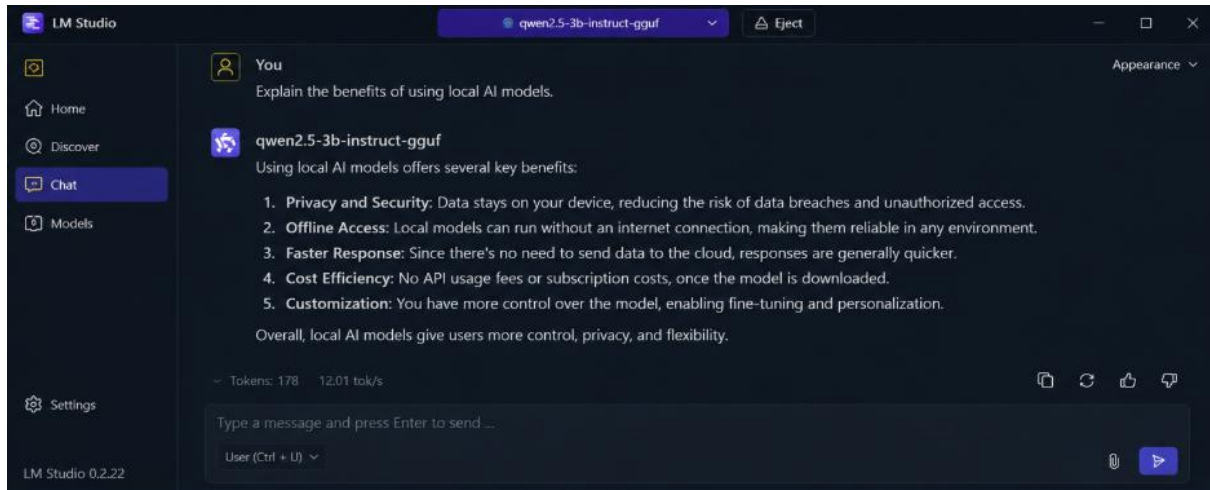
Gemma 3 1B is suitable for computers with limited resources and offers quick responses. Qwen 2.5 3B provides better response quality and understanding but requires more memory and processing power.

Conclusion:

Both models performed well. Gemma 3 1B was faster and lightweight, whereas Qwen 2.5 3B produced higher-quality and more detailed responses. The choice depends on the user's hardware and requirements.

Reflection Questions

Screenshot(s):



Reflection Questions